

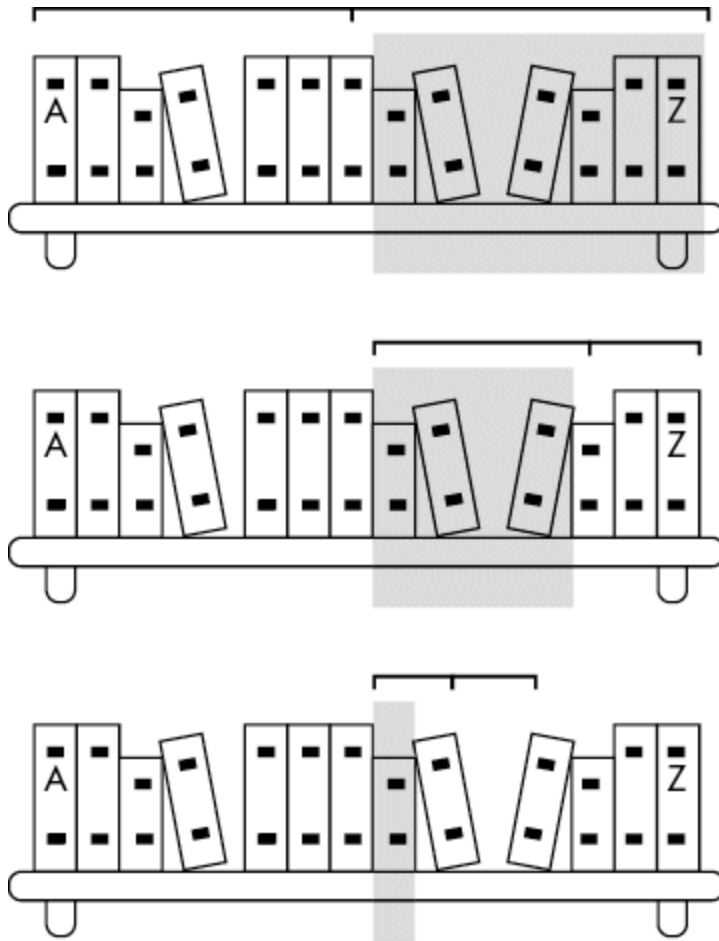


Figure 5-1: A binary search repeatedly determines which half of a range contains your target item in a sorted array of items.

This process scales efficiently; doubling the number of books to search adds only one step to the search process. A linear search of a shelf with 50 books takes 50 steps, and a linear search of a shelf with 100 books takes 100 steps. But a binary search of a shelf with 50 books takes only 6 steps, and a shelf with 100 books takes only 7 steps.

Let's ask the three recursion questions about our binary search implementation:

What is the base case? Searching a range of items that is only one item in length.

What argument is passed to the recursive function call? The indices of the left and right ends of the range in the list we are searching.